



User Maintenance Manual

Version 1.0

This document serves as a comprehensive guide for developers, maintenance personnel, and end-users of the PUPCON system. It provides detailed instructions and reference materials to ensure proper usage, effective maintenance, and efficient management of the system's features.

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i Note

This document ONLY provides a brief and basic technical overview of the **PUPSCHEd** project, a faculty scheduling system designed for ease of use and automation. To test and practice LaTeX documentation workflow.

System Overview

Key Features

Feature	Description
Secure Authentication	Implements JWT-based login, session handling, and role-based access control for system security.
Admin Management Tools	Provides full CRUD operations for users, courses, and schedules, including system configuration.
Faculty Access	Read-only access to schedules with personal dashboards for monitoring teaching loads.
Responsive Web Design	Mobile-friendly interface optimized for cross-device compatibility and performance.
Real-time Updates	Supports live schedule modifications with immediate data synchronization.
Data Integrity	Utilizes a PostgreSQL-backed database ensuring consistency, reliability, and transactional safety.

Table 1: Key technical features of the PUPSCHEd system.

Architecture Diagram

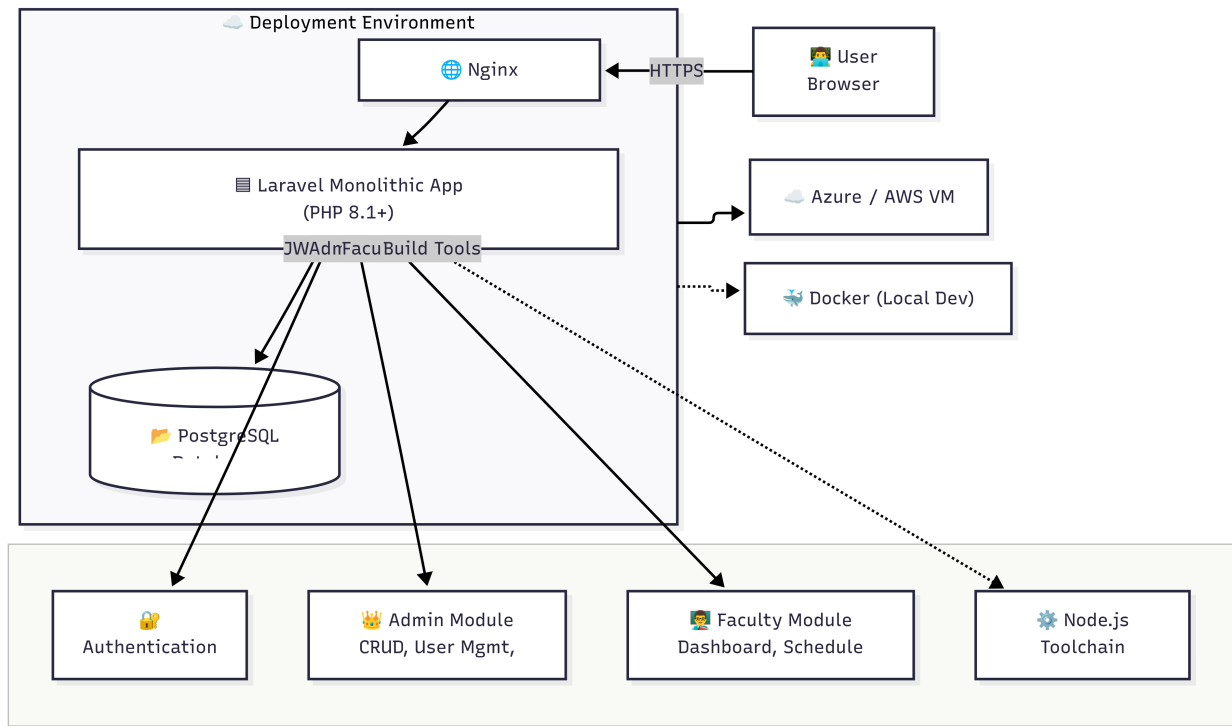


Figure 1: System Architecture of PUPSCHEd.

Development Setup

Installation Requirements

Package	Version Requirement
PHP	≥ 8.1
Node.js	$\geq 16.x$
PostgreSQL	$\geq 13.x$
Composer	Latest
Docker	Latest

Table 2: Required packages for the development environment.

Step-by-Step Guide

Step 1 – Clone Repository

```
# HTTPS
git clone https://github.com/1101101011/PUPSCHEd.git

# SSH
git clone git@github.com:1101101011/PUPSCHEd.git
```

Step 2 – Navigate to Project

```
cd PUPSCHEd
```

Step 3 – Install Dependencies

```
# Frontend dependencies
npm install

# Backend dependencies
composer install
```

Step 4 – Environment Setup

```
# Copy environment file
cp .env.example .env

# Generate application key
php artisan key:generate
```

Step 5 – Database Configuration

```
# Update .env file with your credentials
DB_USERNAME=your-username
DB_PORT=5432
```

Step 6 – Database Migration

```
# Run migrations
php artisan migrate

# Fresh migration (optional)
php artisan migrate:fresh

# Seed database
php artisan db:seed
```

Step 7 – Start Development Server

```
composer run dev
```